UNIVERSITY OF PERADENIYA

DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE

END SEMESTER EXAMINATION - SEMESTER I (2020/2021)

CSC1041 - PROGRAMMING LABORATORY I

Answer **ALL** Questions Time Allowed: **Three** Hours

PART A - (C PROGRAMMING)

Save your C program as S19XXXQ1.c where S19XXX is your registration number. Extra marks will be awarded for using good programming practices.

1. [50 Marks]

Create a C program to order pizza from a pizza store. The program should collect user data including the customer name, nic, mobile, pizza types, topping details, and quantities. A customer can buy a maximum of 10 pizzas at a time. Available Pizza types and relevant prices for the regular size pizzas are as follows. For each Large size pizza an extra 250 rupees per item must be paid at the checkout. *

Pizza Type	**Hot & Spicy**	**Cheesy**	**Veggie**	**Meat**	**BBQ Chicken**
Price (Rs.) [Regular Size]	500.00	650.00	500.00	700.00	750.00

Topping type	**Pepperoni**	**Mushroom**	**Extra cheese**	**Onion**	**Green pepper**
Price (Rs.)	100.00	150.00	200.00	150.00	100.00

A recurring menu should be implemented in order to place orders for different types of pizza. There should be separate functions to print the pizza price list and the toppings price list. After placing an order, a receipt should be printed as below with the purchase summary;

```
Example output:
----------Oder Details---------
       Customer name  : Kamal
       Customer ID    : 12345678912v
       Customer Phone : 1234566666

       Regular      Hot & Spicy  - Onion                  3 →  1950.00
       Large        BBQ Chicken  - Pepperoni              1 →  1100.00
       Large        Meat         - Mushroom               4 →  4400.00
                                                      TOTAL →  7450.00
---------Thank you------
```

The expected menu functionality should be similar to the below example.

```
Example:
------Pizza Store Menu------
         1.Order Pizza.
         2.View Pizza Menu.
         3.View Toppings Menu.
         9.Exit.
Your Selection: 1
insert national id number : 1234567v
insert name: Kamal
insert mobile number: 0771233456
--------Pizza Prices-------
1. hot & spicy     500.00
2. cheesy          650.00
3. veggie          500.00
4. Meat            700.00
5. BBQ Chicken     750.00
---------------------------
Select your choice: 3
--------Topping Prices-------
1. Pepperoni       100.00
2. Mushroom        150.00
3. Extra cheese    200.00
4. Onion           150.00
5. Green pepper    100.00
---------------------------
Select your choice: 2
input pizza quantity: 2
-----select pizza size------
 1.Regular
 2.Large
 Select your choice: 2
```

PART B - (PYTHON PROGRAMMING)

Create a folder named CSC1041_End_S19XXX in your H drive. Copy the shared text files into the newly created folder. Save your Python programs inside the same folder as S19XXXQ2.py, S19XXXQ3.py, S19XXXQ4.py for each question where S19XXX is your registration number.

2. [30 Marks]

A. In programming String manipulation refers to the process of handling and analyzing strings. It involves various operations concerned with modification and parsing of strings to use and change its data. Write a function [***get_counts()***] to get the below counts from the file ("text_file.txt") available to you in your home directory. You may consider all the words separated by other than blank spaces as single words.

i. Number of words starting with UPPER-case letters
ii. Number of words only having LOWER-case letters
iii. Number of words starting with character A or a
iv. Number of words having character R or r as the second last character

B. Update the above function so that it groups the words into 4 different sequences according to the above 4 categories. Furthermore, the categorized words should not have below four any of the symbols given below as leading or ending characters.

.	,	(	)	"	"

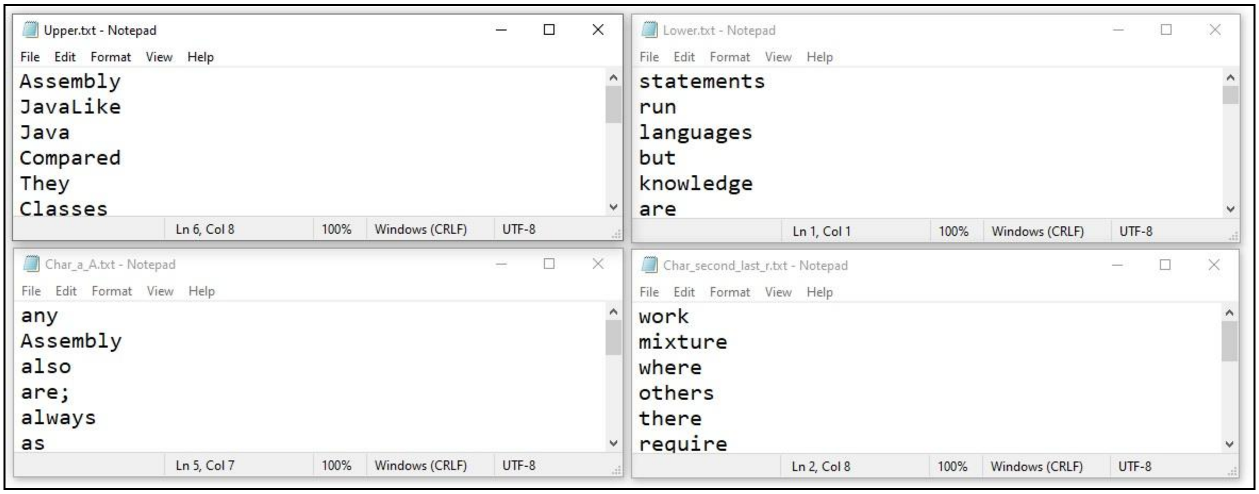
C. Write the above separated word sequences into four different text files as follows. **When writing the content**, you should remove the duplicate words while keeping their first occurrence inside these text files. Also, each word should be written in a new line in the text file.

- Upper.txt
- Lower.txt
- Char_a_A.txt
- Char_second_last_r.txt

```
Example output:

Words starting with UPPER-case letters                          : 22
Words starting with LOWER-case letters                          : 231
Words starting with character A or a                            : 42
Words having character R or r as the second last character      : 23

upper               :  'Java', 'Compared', 'Figures', 'Hello', 'World', …
lower               :  'is', 'one', 'of', 'many', 'high-level', 'programming', …
char_a_A            :  'abstraction', 'and', 'and', 'a', 'any','an','a', 'Assembly',
…
char_second_last_r  :  'software', 'computers', 'mixture', 'form', 'are', …
```



3. **[12 Marks]**
Create a Python function that takes a string as a user input and prints the letter frequencies. The function should convert all the input letters to lowercase and only count the letters from ‘a’ to ‘z’.

```
Example output:

Enter the text: Hello World Good Morning
Converted to Lower-case : hello world good morning
Letter Frequencies :
{'h': 1, 'e': 1, 'l': 3, 'o': 5, 'w': 1, 'r': 2, 'd': 2, 'g': 2, 'm': 1, 'n': 2, 'i': 1}
```

4. **[08 Marks]**
Create a Python function that takes two strings as user inputs and prints the common words available in those two strings. If the two strings contain the same word more than once, it should only print a single occurrence of that particular word. (You may ignore the case sensitivity)

```
Example output:

Enter the text 1: Hello Good Morning People of Sri Lanka
Enter the text 2: Hello Good Evening People of Sri Jayewardenepura

Common Words
['hello', 'of', 'sri', 'good', 'people']
```